

SAT Math Mastery · Week 1B**Lines, Slope & Linear Functions**

Time	150 minutes: 110 min solving + 40 min self-check and error log
Topics	slope from points/graphs/tables · forms of a line · intercepts · parallel & perpendicular · translations · meaning of slope/intercept · linear function notation
Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Find slope from two points, a graph, a table, a standard-form equation, or a verbal rate.
- S2.** Write and convert between slope-intercept, point-slope, standard and intercept forms.
- S3.** Find x - and y -intercepts and use them (areas, conditions, context meaning).
- S4.** Parallel and perpendicular lines, including directly from $Ax + By = C$.
- S5.** Translate a line (up/down/left/right) and track slope and intercepts.
- S6.** Interpret slope and intercept in context; build $f(x) = mx + b$ from data and solve for inputs.
- S7.** Coordinate reasoning: collinearity, midpoint, perpendicular bisector, triangles cut by axes.
- S8.** Parameter conditions on lines (a constant fixes slope, intercept, perpendicularity or area).

2 · REQUIRED FACTS

- Slope = $\frac{\text{rise}}{\text{run}}$ = constant rate of change; the y -intercept is the value at $x = 0$; the x -intercept is where $y = 0$.
- Horizontal line $y = c$: slope 0. Vertical line $x = c$: slope undefined.
- Parallel $\iff$ equal slopes (different intercepts). Perpendicular $\iff$ slopes are negative reciprocals (product -1).
- A function is linear if equal steps in x give equal steps in y (check every gap in a table).
- Shifting a graph up k adds k to y ; slope never changes under a translation.
- Two points determine a line; on a graph read only points that sit exactly on grid intersections.

3 · REQUIRED FORMULAS

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad y = mx + b, \quad y - y_1 = m(x - x_1), \quad \frac{x}{a} + \frac{y}{b} = 1$$

$$Ax + By = C: \quad m = -\frac{A}{B}, \quad x\text{-int } \frac{C}{A}, \quad y\text{-int } \frac{C}{B}, \quad \text{area with axes} = \frac{1}{2} \left| \frac{C}{A} \cdot \frac{C}{B} \right|$$

$$\parallel: A_1B_2 = A_2B_1, \quad \perp: A_1A_2 + B_1B_2 = 0, \quad \text{midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right), \quad f(a + d) - f(a) = md$$

4 · MINI THEORY SHEET

Choose the form by the data.

Given	Fastest route
two points	$m = \frac{\Delta y}{\Delta x}$, then point-slope with either point
slope and y -intercept	$y = mx + b$ directly
both intercepts $(a, 0), (0, b)$	$\frac{x}{a} + \frac{y}{b} = 1$, or $m = -\frac{b}{a}$
$Ax + By = C$	read $m = -\frac{A}{B}$; set $x = 0$ or $y = 0$ for intercepts
a table	m from any two rows; confirm the rest have the same m

Perpendicular in standard form. Swap the coefficients and negate one: $\perp$ to $Ax + By = C$ is $Bx - Ay = C'$; find C' from the given point.

Parallel in standard form. Keep A, B (or scale both), change only C .

Translation. Down k : $b \rightarrow b - k$, and the x -intercept moves by $\frac{k}{m}$. Same slope, always.

Meaning. In $y = mx + b$: m = change in y per 1 unit of x ; b = starting value. Units of m are (units of y)/(units of x).

Constant conditions. Turn the condition (slope, perpendicular, area, collinear) into one equation in the constant, then solve.

Whole-number contexts. Solve $f(x) = \text{target}$; if x must be a whole day/trip/session, round to the first value that *reaches* the target.

Speed shortcuts.

Situation	Shortcut
equal input steps	$f(a + d) - f(a) = md$ (no equation needed)
perpendicular, standard form	$Ax + By = C \rightarrow Bx - Ay = C'$, find C' from the point
translate right h / up k	replace x by $x - h$; add k to y
triangle with axes	area = $\frac{C^2}{2 AB }$ for $Ax + By = C$
perpendicular bisector	midpoint + negative reciprocal of the segment's slope
sign of constants	$m = -\frac{A}{B}$, y -int $\frac{C}{B}$, x -int $\frac{C}{A}$: signs fix the quadrants
intersection with a parameter	solve for x first; the parameter may cancel
infinitely many / no solutions	scale one equation: same ratios $\Rightarrow$ same line

5 · GUIDED EXAMPLES

Example 1 — Two points $\rightarrow$ equation $\rightarrow$ intercepts

A line passes through $(-1, 7)$ and $(3, -1)$. Find its equation and both intercepts.

$$m = \frac{-1 - 7}{3 - (-1)} = \frac{-8}{4} = -2. \text{ Point-slope: } y - 7 = -2(x + 1) \Rightarrow y = -2x + 5.$$

$$y\text{-intercept } 5; x\text{-intercept: } 0 = -2x + 5 \Rightarrow x = \frac{5}{2}.$$

Example 2 — Perpendicular through a point (standard form)

Line ℓ : $4x - 3y = 12$. Write the line through $(2, -1)$ perpendicular to ℓ .

Swap and negate: $3x + 4y = C'$. Plug in $(2, -1)$: $C' = 3(2) + 4(-1) = 2$. Answer: $3x + 4y = 2$.

Example 3 — Meaning of slope and intercept

A gym plan costs \$340 after 2 months and \$640 after 5 months (linear). Write $C(m)$ and interpret its parts.

$$\text{slope} = \frac{640 - 340}{5 - 2} = 100. \quad C(m) = 100m + b; \quad 340 = 200 + b \Rightarrow b = 140. \quad \text{So } C(m) = 100m + 140.$$

100 = dollars per month; 140 = joining fee (cost at $m = 0$).

6 · PRACTICE SET — 40 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

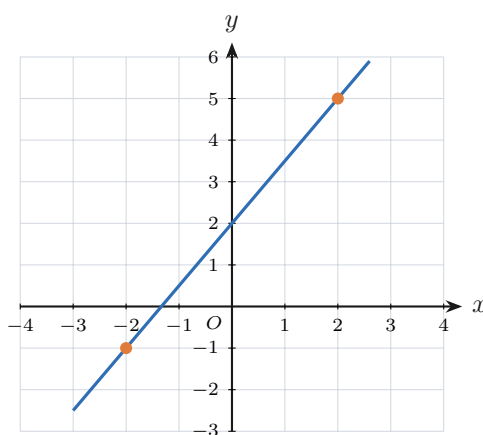
EASY · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 7x - 4$. What is the value of $f(3) + f(-1)$?

2.

EASY · DESMOS

The graph of a line in the xy -plane is shown. Which equation represents the line?



(A) $y = -\frac{3}{2}x + 2$

(B) $y = \frac{3}{2}x + 2$

(C) $y = \frac{2}{3}x + 2$

(D) $y = \frac{3}{2}x - 2$

3.

EASY · NO CALCULATOR NEEDED

What is the x -intercept of the line given by $3x + 4y = 24$?

(A) $(0, 8)$

(B) $(6, 0)$

(C) $(8, 0)$

(D) $(0, 6)$

4.

EASY · CALCULATOR

A tank contains 90 liters of water after 2 minutes of draining and 66 liters after 6 minutes. The water level decreases at a constant rate. What is the rate of change of the volume, in liters per minute? (Enter the signed value.)

5.

EASY · NO CALCULATOR NEEDED

Which of the following lines is perpendicular to the line $2x + y = 7$?

(A) $x - 2y = 4$

(B) $2x - y = 3$

(C) $x + 2y = 1$

(D) $2x + y = -3$

6.

MEDIUM · NO CALCULATOR NEEDED

The function f is linear. Selected values are shown in the table. What is the value of p ?

x	$f(x)$
1	7
4	19
6	p

(A) 31

(B) 21

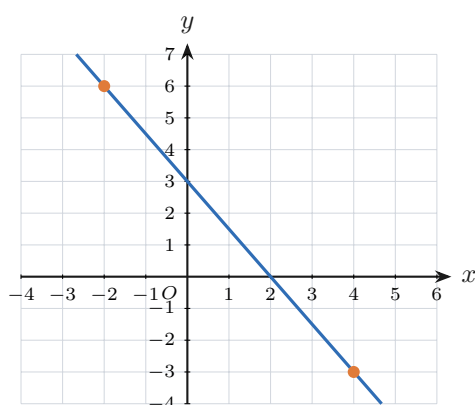
(C) 23

(D) 27

7.

MEDIUM · DESMOS

The graph of a line is shown. Which equation represents it?



(A) $3x - 2y = 6$

(B) $3x + 2y = -6$

(C) $3x + 2y = 6$

(D) $2x + 3y = 6$

8.

MEDIUM · NO CALCULATOR NEEDED

Which of the following is an equation of the line that passes through $(-4, 3)$ and is perpendicular to the line $5x - 2y = 9$?

(A) $5x + 2y = -14$

(B) $2x + 5y = 7$

(C) $2x - 5y = -23$

(D) $5x - 2y = -26$

9.

MEDIUM · NO CALCULATOR NEEDED

The line $y = -2x + 6$ is translated 3 units to the right. Which equation defines the new line?

(A) $y = -2x$

(B) $y = -2x + 9$

(C) $y = -2x + 3$

(D) $y = -2x + 12$

10.

MEDIUM · NO CALCULATOR NEEDED

The altitude, in meters, of a drone t seconds after takeoff is modeled by $h(t) = 45 + 12t$ for $0 \leq t \leq 20$. Which of the following is the best interpretation of $h(5) = 105$?

(A) At 5 s, the altitude is 105 m.

(B) The drone climbs 105 m every 5 s.

(C) The drone starts at 105 m.

(D) The drone climbs at 5 m/s.

11.

MEDIUM · CALCULATOR

A pool holds 140 liters of water and is filled at a constant rate of 35 liters per minute. What is the least whole number of minutes after which the pool holds at least 1000 liters?

12.

MEDIUM · NO CALCULATOR NEEDED

The points $(1, a)$, $(3, 5)$ and $(7, 11)$ lie on the same line. What is the value of a ?

(A) -2 (B) 2 (C) 3 (D) $\frac{7}{2}$ **13.**

MEDIUM · NO CALCULATOR NEEDED

A line passes through the points $(-3, 5)$ and $(9, -1)$. What is the x -coordinate of its x -intercept?

(A) -7 (B) $\frac{7}{2}$ (C) 7 (D) 14 **14.**

HARD · NO CALCULATOR NEEDED

The function f is linear and $f(t + 6) - f(t) = 21$ for every value of t . If $f(-2) = 3$, what is the value of $f(8)$?

15.

HARD · NO CALCULATOR NEEDED

If $f(x) = 4x - 3$ and $f(2a) = 3f(a) + 7$, what is the value of a ?

16.

HARD · NO CALCULATOR NEEDED

$$ax + 3y = 12$$

$$4x + 6y = b$$

In the system above, a and b are constants. If the system has infinitely many solutions, what is the value of $a + b$?

17.

HARD · NO CALCULATOR NEEDED

In the xy -plane, the line $kx + 3y = 7$ is perpendicular to the line $2x - 5y = 1$, where k is a constant. What is the value of k ?

(A) $\frac{15}{2}$

(B) $\frac{6}{5}$

(C) $-\frac{15}{2}$

(D) $\frac{2}{15}$

18.

HARD · NO CALCULATOR NEEDED

In the xy -plane, the graph of $Ax + By = C$ is a line, where $A > 0$, $B < 0$ and $C < 0$. Which quadrant does the line *not* pass through?

- (A) Quadrant I (B) Quadrant II
(C) Quadrant III (D) Quadrant IV

19.

HARD · NO CALCULATOR NEEDED

Points $A(-2, 9)$ and $B(6, -3)$ are the endpoints of a segment. Which of the following is an equation of the perpendicular bisector of $\overline{AB}$?

- (A) $3x - 2y = 0$ (B) $2x + 3y = 13$
(C) $2x - 3y = -5$ (D) $3x + 2y = 12$

20.

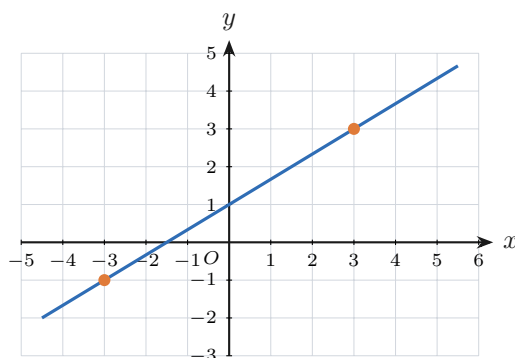
HARD · NO CALCULATOR NEEDED

A line with slope -2 passes through the point $(3, 4)$. The line and the positive coordinate axes bound a triangle. What is the area of the triangle?

21.

HARD · DESMOS

The graph of line ℓ is shown. Line m is perpendicular to ℓ and passes through the point $(4, 1)$. What is the y -intercept of line m ?



22.

HARD · CALCULATOR

Company A charges a \$40 setup fee plus \$12 per hour for a rental. Company B charges \$15 per hour with no setup fee. What is the least whole number of hours for which Company A's total cost is less than Company B's?

23.

HARD · NO CALCULATOR NEEDED

In an arithmetic sequence, the 4th term is -5 and the 12th term is 27. What is the 20th term?

(A) 55

(B) 59

(C) 63

(D) 51

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a positive constant K , the line $3x + 4y = K$ and the two coordinate axes bound a triangle of area 54. What is the value of K ?

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, line k has slope $\frac{2}{3}$ and passes through $(6, -1)$. Line m is perpendicular to line k and has the same y -intercept as line k . What is the x -coordinate of the x -intercept of line m ?

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

The line $y = 2x + 4$ crosses the x -axis at point A and the y -axis at point B . Line m passes through A and is perpendicular to $y = 2x + 4$. Line m crosses the y -axis at point C . What is the area of triangle ABC ?

27.

SAT CHALLENGE · DESMOS

In the xy -plane, line p is $y = kx + 4$ and line q is $y = (k + 2)x - 6$, where k is a constant. The two lines intersect at a point whose y -coordinate is 19. What is the value of k ?

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

The linear functions f and g are defined by $f(x) = 2x + 1$ and $g(x) = ax + b$, where a and b are constants. If $f(g(x)) = 6x - 5$ for all x , what is the value of $g(f(2))$?

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

A line has x -intercept a and y -intercept b , where $a \neq 0$ and $b = 2a$. If the line passes through $(3, 5)$, what is the value of b ?

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

For linear function p , $p(c) = -6$, $p(5) = 42$, and the slope of the graph of p is 8. For linear function t , $t(c) = -7$ and $t(6) = 56$. What is the slope of the graph of t ?

31.

SAT CHALLENGE · NO CALCULATOR NEEDED

Line p has equation $3x - ky = 15$, where k is a nonzero constant. Line q is perpendicular to line p and passes through the y -intercept of line p . Which of the following is an equation of line q ?

(A) $y = -\frac{k}{3}x + \frac{15}{k}$

(B) $y = \frac{k}{3}x - \frac{15}{k}$

(C) $y = -\frac{3}{k}x - \frac{15}{k}$

(D) $y = -\frac{k}{3}x - \frac{15}{k}$

32.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, line ℓ has equation $5x - 2y = 11$ and line m has equation $ax + by = d$, where a , b and d are positive constants and $a + b = 21$. Lines ℓ and m are perpendicular, and line m passes through the point $(4, 3)$. What is the value of d ?

33.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is linear. The function g is defined by $g(x) = -2f(x - 1) + 5$, and the graph of $y = g(x)$ passes through the points $(0, 3)$ and $(2, -5)$. The graph of $y = h(x)$ is the graph of $y = f(x)$ translated 2 units to the left and 6 units up. What is the y -coordinate of the y -intercept of the graph of h ?

34.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is linear. The function g is defined by $g(x) = \frac{f(x+2) - x}{x-3}$ for $x \neq 3$. Some values of g are shown in the table below.

x	1	5
$g(x)$	3	-11

What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

(A) 4

(B) -3

(C) -5

(D) -4

35.

SAT CHALLENGE · NO CALCULATOR NEEDED

The linear function f satisfies $f\left(\frac{2}{3}\right) = \frac{1}{4}$ and $f\left(\frac{8}{3}\right) = \frac{13}{4}$. What is the value of $f\left(\frac{1}{2} + \frac{3}{5}\right)$?

36.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$kx + 3y = 12$$

$$4x + (k - 1)y = 12$$

In the system of equations above, k is a constant. For which value of k does the system have no solution?

(A) -4 (B) -3 (C) 4 (D) 3

37.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, a line with negative slope passes through the point $(3, 4)$. The line and the positive coordinate axes bound a triangle of area 24. What is the slope of the line?

38.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the lines $kx + y = 7$ and $x - ky = 3$ are graphed, where k is a real constant. For which values of k are the two lines perpendicular?

(A) $k = 1$ only(B) $k = 1$ and $k = -1$ only(C) All real values of k (D) $k = 0$ only**39.**

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is linear, and $f(5) - f(1) = 12$. The function g is defined by $g(x) = f(3x) + 1$. What is the value of $g(2) - g(0)$?

40.

SAT CHALLENGE · NO CALCULATOR NEEDED

The line $y = 2x + 3$ is reflected across the y -axis and then translated 4 units down. Which equation defines the resulting line?

(A) $y = -2x - 1$

(B) $y = -2x + 7$

(C) $y = 2x - 1$

(D) $y = -\frac{1}{2}x - 1$